



DAWOOD UNIVERSITY OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF BASIC SCIENCES AND HUMANITIES

Subject:	Applied Physics	
Course code:	BS-1001	
Discipline:	Bachelor of Cyber Security	
Semester:	First Semester, First year	
Effectiveness:	Batch-2023	
Course type:	Compulsory	
Marks:	Theory: 100	Practical: 50
Credit Hours:	3 CH	1CH
Teaching Scheme:	3HRS/week	3HRS/week
Assessment:	20% sessional, 30% Mid semester, 50% Final Exam	

Books:

1. physics For scientists and engineers by Serway Jewett
2. Semi conductor Physics by Macalve
3. University physics by Freeman and Young

Waves



What Are Waves?

- A wave is a disturbance that carries energy through matter or space without transferring matter.
- Wave Demo

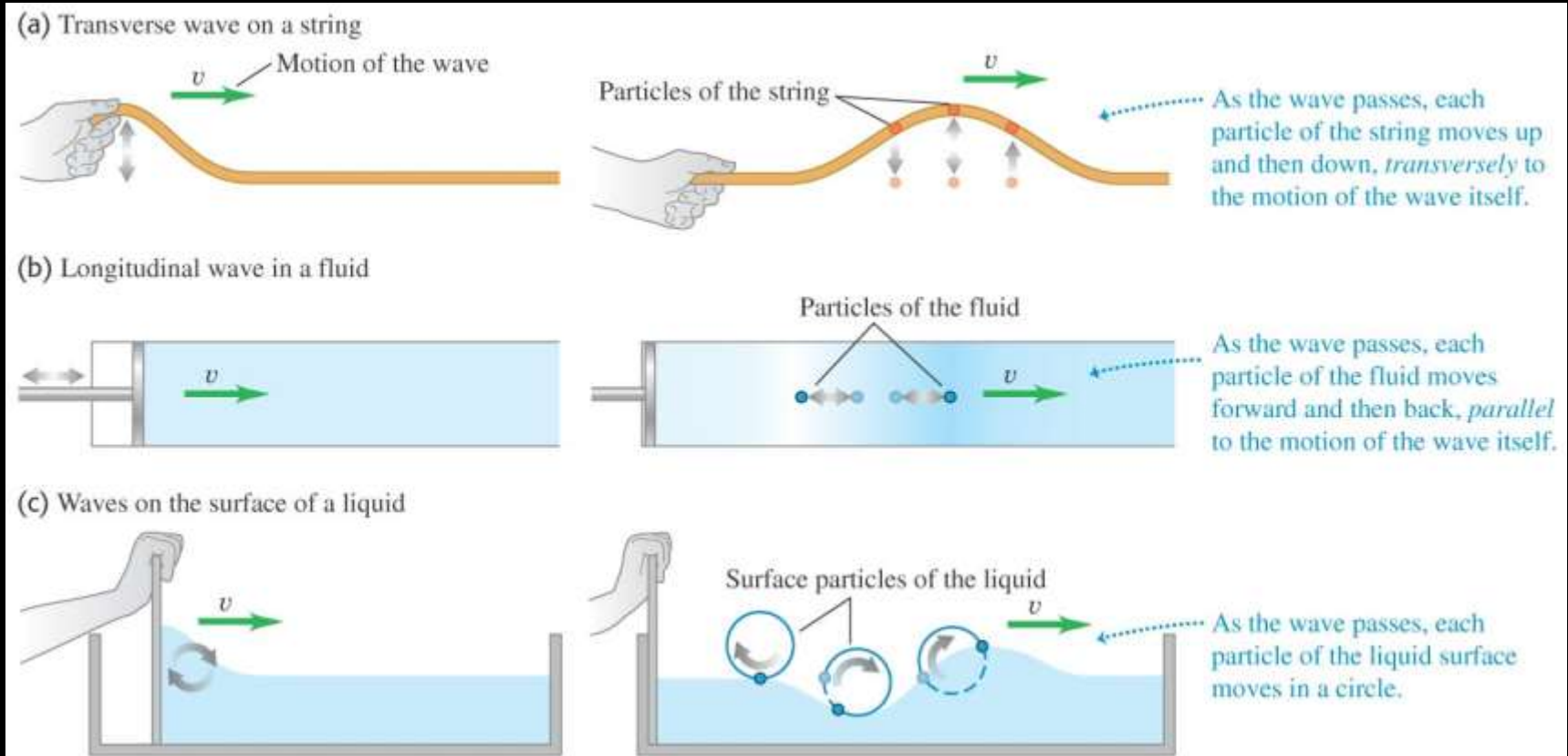


Types of Waves

1. **Mechanical waves.** These waves have two central features: They are governed by Newton's laws, and they can exist only within a material medium, such as water, air, and rock. Common examples include water waves, sound waves, and seismic waves.
2. **Electromagnetic waves.** These waves require no material medium to exist. All electromagnetic waves travel through a vacuum at the same exact speed $c = 299,792,458 \text{ m/s}$. Common examples include visible and ultraviolet light, radio and television waves, microwaves, x rays, and radar.
3. **Matter waves.** These waves are associated with electrons, protons, and other fundamental particles, and even atoms and molecules. These waves are also called matter waves.

Types of mechanical waves

- A mechanical wave is a disturbance traveling through a medium.
- Figure below illustrates transverse waves and longitudinal waves.

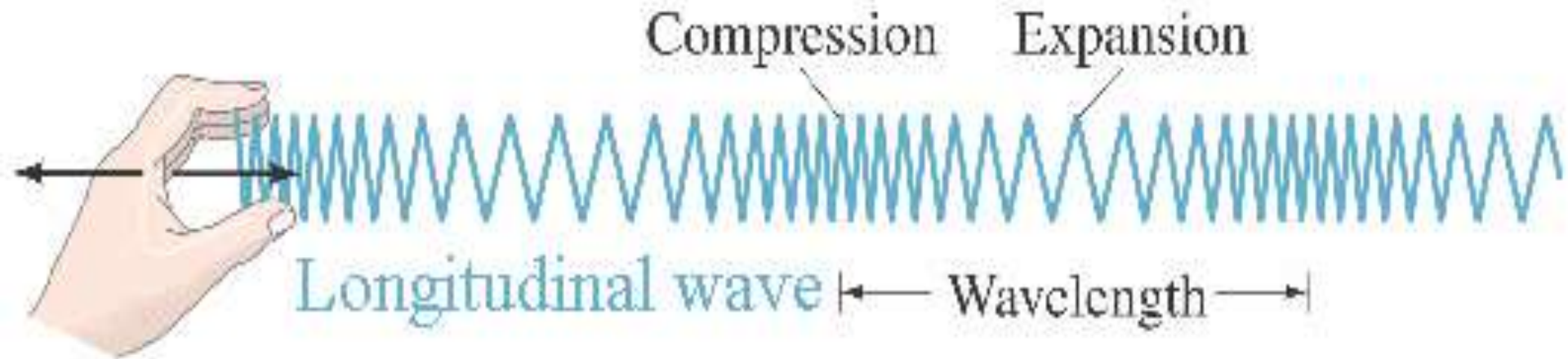
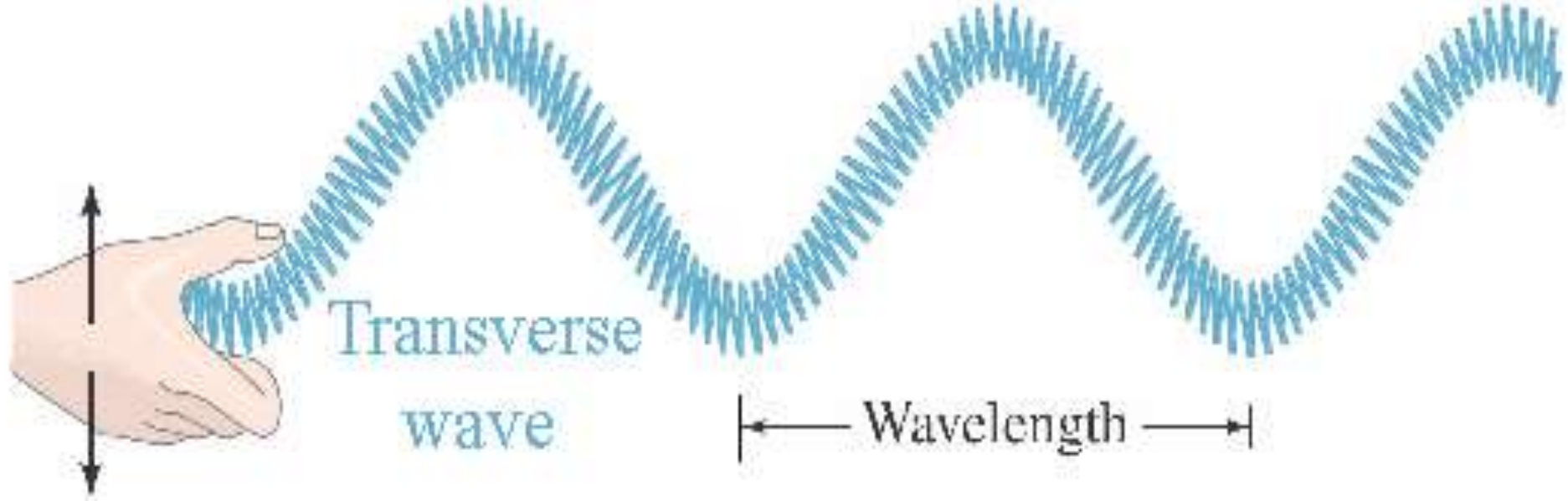


Types of mechanical waves

- There are two types of waves!

1. Transverse Waves: A wave that disturbs the particles in the medium perpendicular to the direction of the wave's travel.
2. Longitudinal Waves: A wave that disturbs the particles in the medium parallel to the direction of the wave's travel.

Transverse vs. Longitudinal Waves

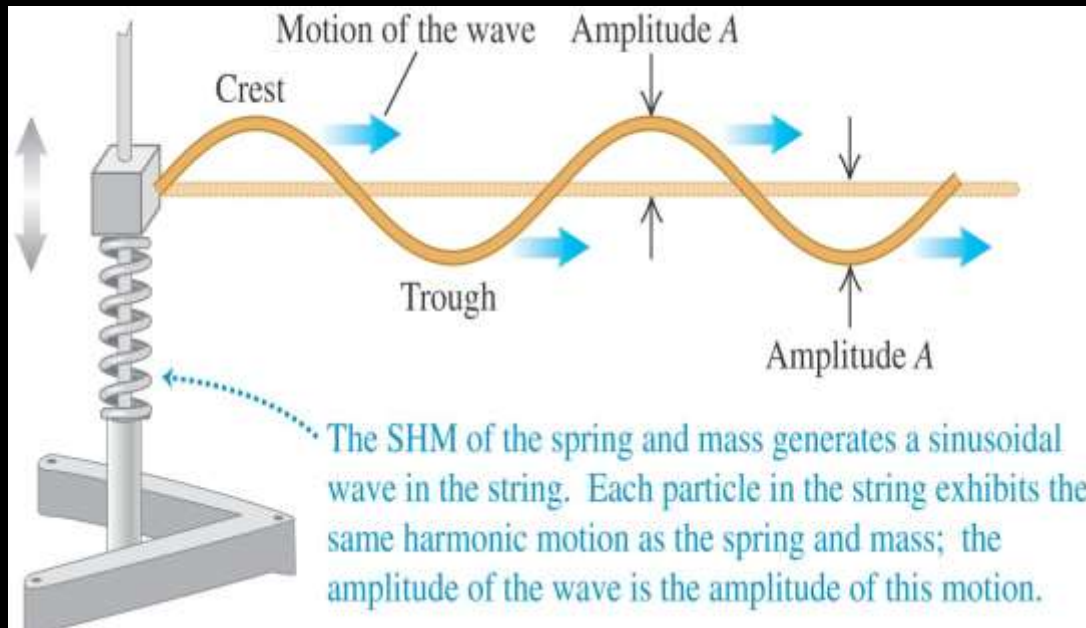


Periodic waves

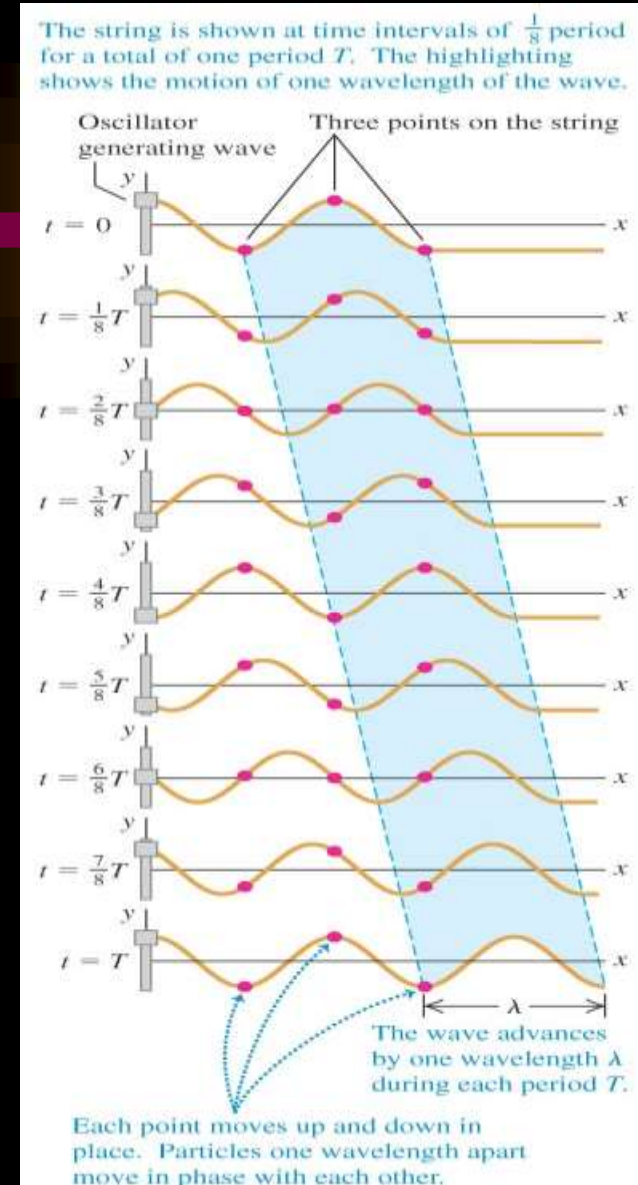
- For a *periodic wave*, each particle of the medium undergoes periodic motion. The speed of the wave is not the same as the speed of the particles.
- The *wavelength* λ of a periodic wave is the length of one complete wave pattern.
- The speed of any periodic wave of frequency f is $v = \lambda f$.
- Example 15.1: The speed of sound in air at 20° C is 344 m/s. What is the wavelength of a sound wave in air at 20° C if the frequency is 262 Hz?

Periodic *transverse* waves

- For the transverse waves shown here in Figures the particles move up and down, but the wave moves to the right.

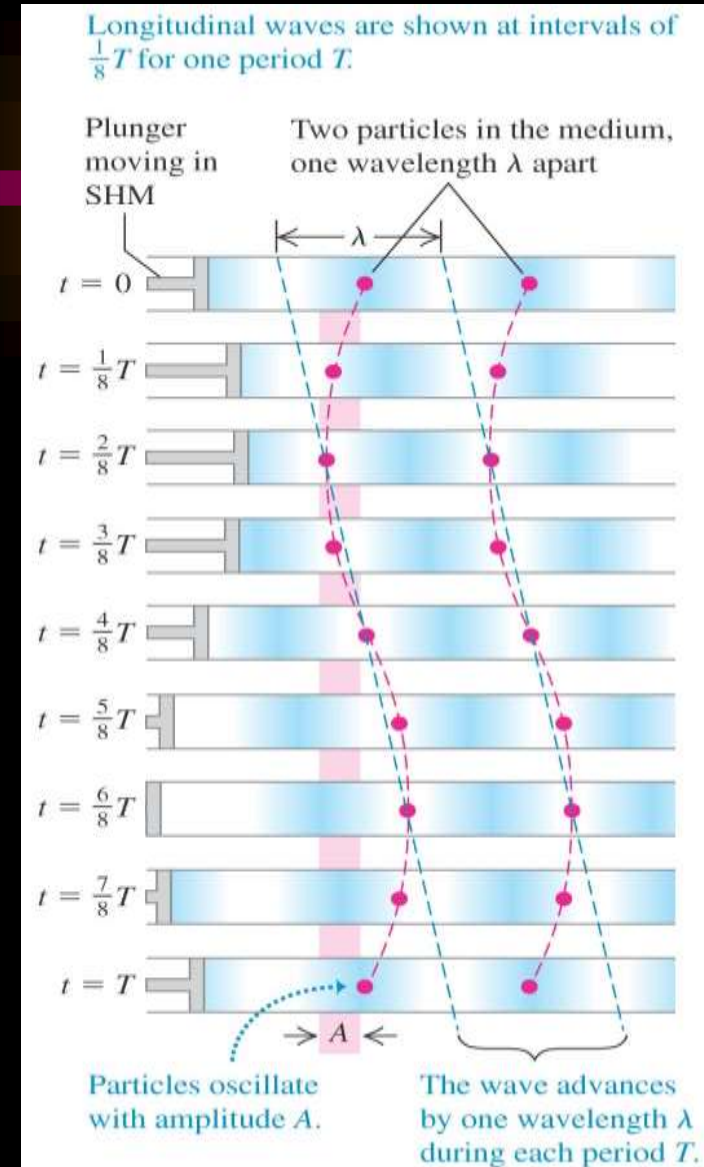
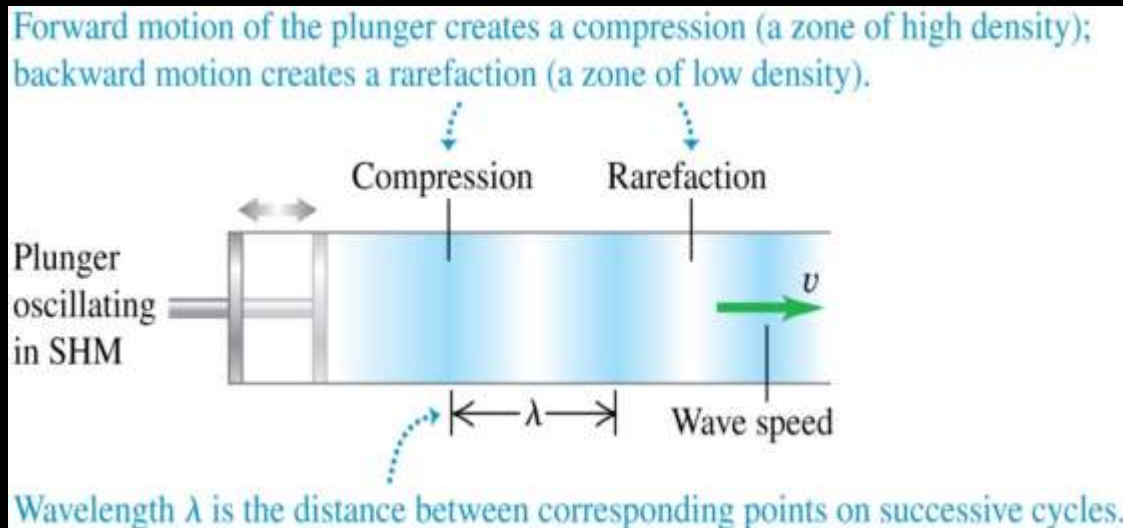


- This difference in direction of the waves and particles is why the wave is called a transverse wave.
- Note that the restoring force is transverse to the direction of the wave propagation.



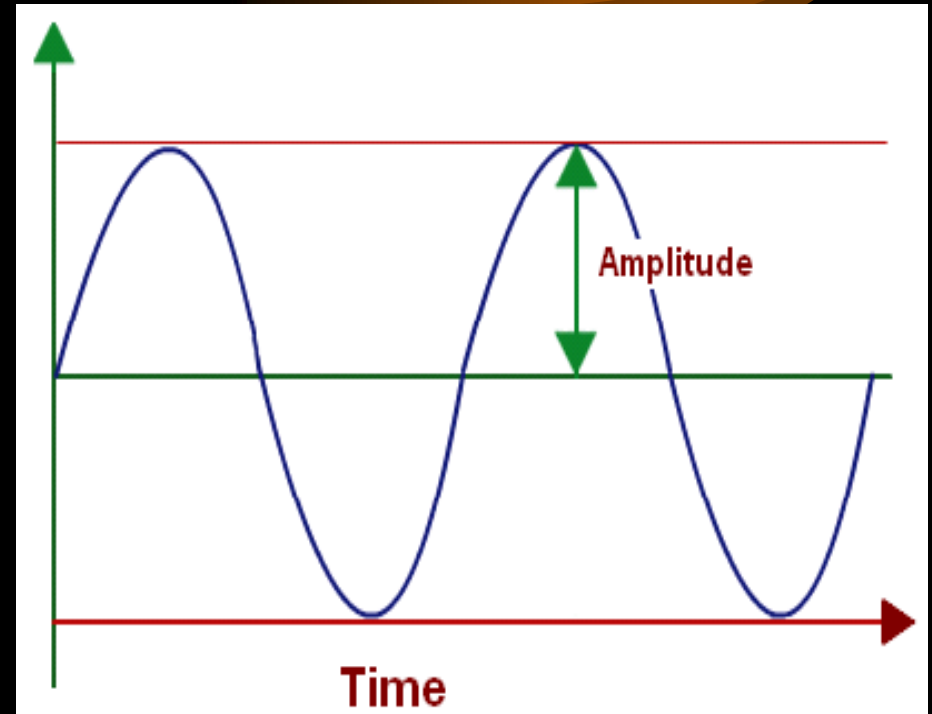
Periodic longitudinal waves

- For the longitudinal waves shown here in Figures. The particles oscillate back and forth along the same direction that the wave moves.
- The restoring force (pressure) is in the same direction as the wave propagation.



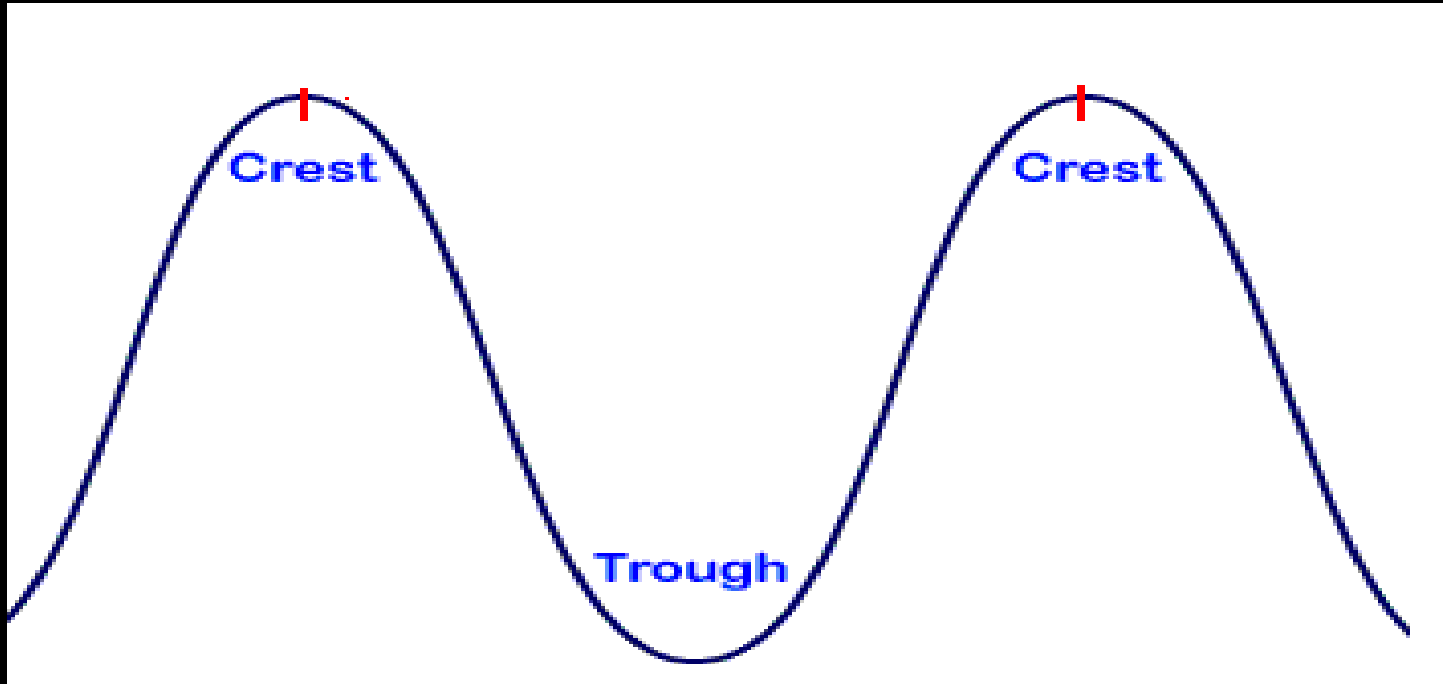
Wave Characteristics

- **Amplitude:** The greatest distance from equilibrium.
- The bigger the amplitude of the wave, the more energy the wave has.



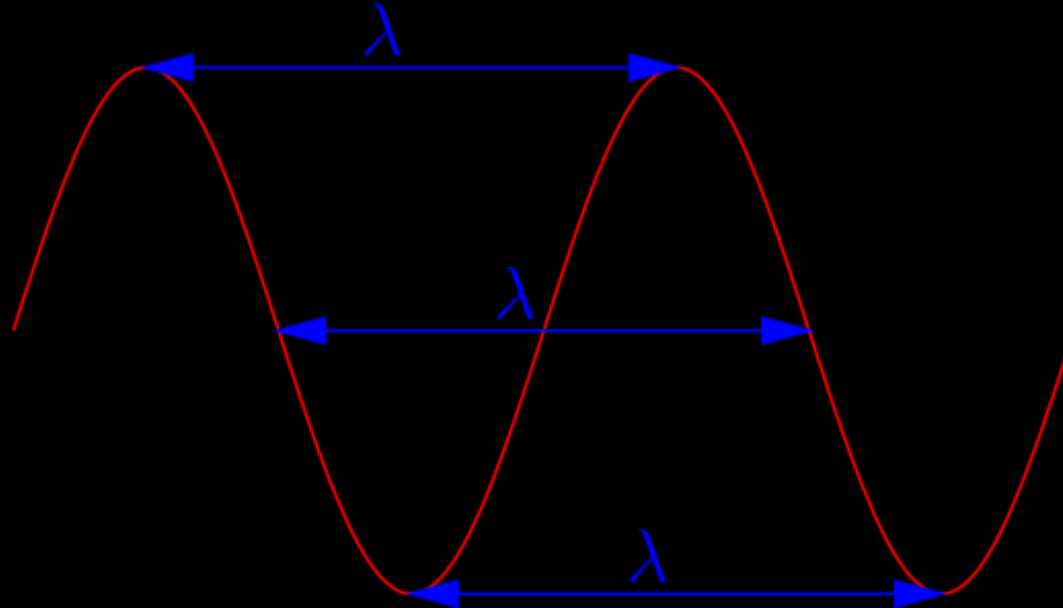
Wave Characteristics

- **Crest:** The upper portion of a wave from mean position.
- **Trough:** The lower portion of a wave from mean position.



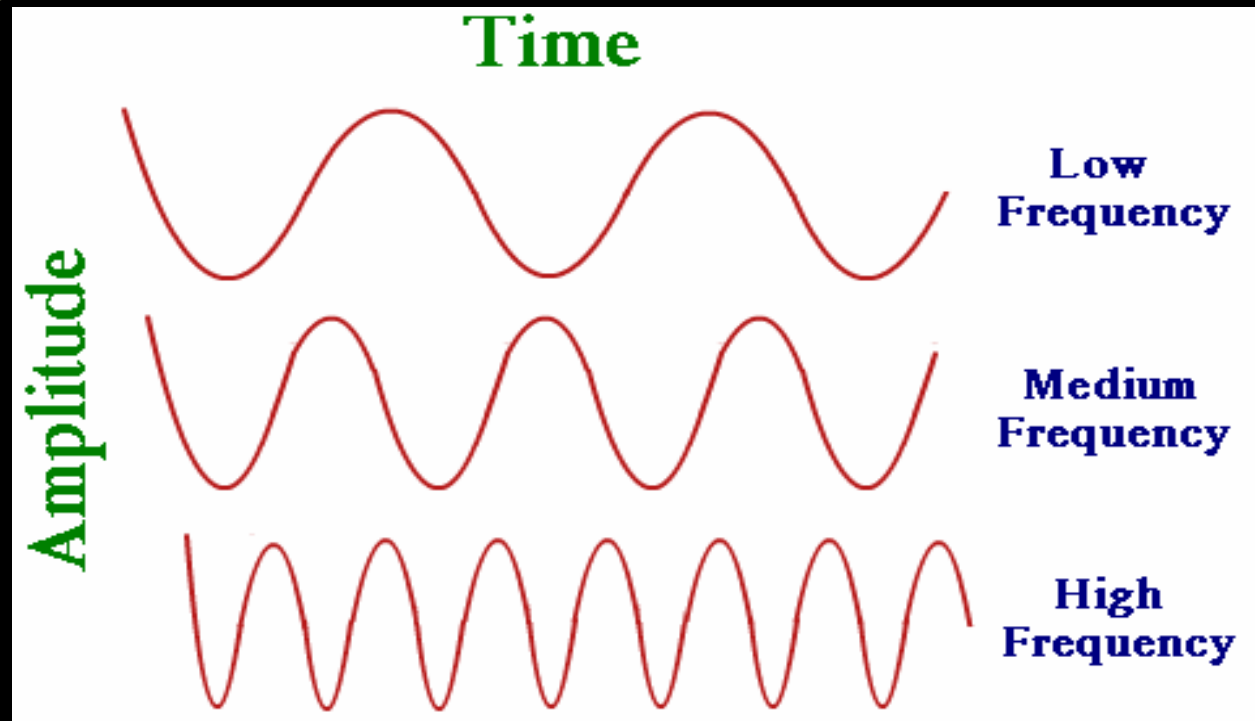
Wave Characteristics

- **Wavelength:** The shortest distance between points where the wave pattern repeats itself.
- Wavelength is measured in meters, the symbol is λ



Wave Characteristics

- **Frequency:** Is the number of complete oscillations a point on that wave makes in each second.
- The frequency is measured in Hertz (Hz) and its unit is $1/s$ or Hz



Wave Characteristics

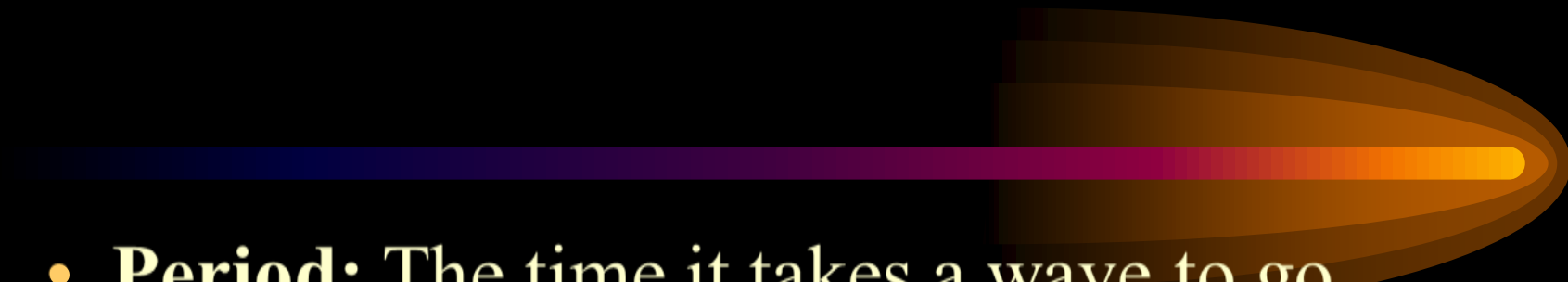
- **Speed:** The speed or velocity of a wave is how fast the energy is moved. For most waves, wave speed does not depend on amplitude, frequency, or wavelength. Speed depends only on the medium through which it moves.

- $v = \frac{m}{s}$

- $\lambda = m, f = \frac{1}{s}$

- $v = f\lambda$

Wave Characteristics



- **Period:** The time it takes a wave to go through one cycle, or the time it takes a point to go through one phase of the wave.
- The period of a wave is measured in seconds, and it's symbol is “T”.
- $T = \frac{1}{f}$

Problem



- What is the velocity of a wave that has a frequency of 250Hz and a wavelength of 1.5m?
- $v = f\lambda$
- $v = 250\text{Hz} * 1.5\text{m} = 375 \frac{\text{m}}{\text{s}}$

Problem

- How does the speed of a wave change if the amplitude is increased?
- The speed doesn't change.
- What is the velocity of a wave that has a frequency of 100Hz and a wavelength of 2.0m?
- $v = f\lambda$
- $v = 100\text{Hz} * 2.0\text{m} = 200 \frac{\text{m}}{\text{s}}$